GOURAV A

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OBJECTIVE

Third-year Computer Science Engineering student working toward a career in machine learning research. Hands-on experience in self-supervised representation learning, mechanistic interpretability, robustness evaluation of vision-language models, and applied full-stack systems. Comfortable building ML systems end to end, from architecture through a deployed interface, with a track record of shipped, award-winning projects.

EDUCATION

Bachelor of Engineering, Computer Science Engineering — APS College of Engineering, Bengaluru 2024 – 2028 (Expected)

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Machine Learning, Operating Systems, Computer Networks

TECHNICAL SKILLS

- **Languages:** Java, Python, C, C++, JavaScript, HTML
- **Web / Frameworks:** React.js, FastAPI, Next.js, Supabase
- **ML / AI:** Machine Learning fundamentals, Deep Learning, self-supervised representation learning, mechanistic interpretability, Edge AI, IoT-based sensing systems, multi-agent systems
- **Core CS:** Data Structures & Algorithms (250+ problems solved, daily practice), OOP, DBMS, Git/GitHub
- **Tools:** VS Code, GitHub, Figma, Docker

CERTIFICATIONS

- **Oracle AI Foundations Associate** — Oracle, Issued Jan 2026
- **GenAI Powered Data Analytics Job Simulation** — Tata Group (Forage), Issued Oct 2026

EXPERIENCE

AI Engineering Intern — Infosys *June 2026 – Present*

- Building an AI Academic Project Mentor platform end to end on Next.js, FastAPI, and Supabase (PostgreSQL, authentication, row-level security), from data schema design through a shipped onboarding UI.
- Completed Milestones 1–3 with the team: system and ER diagrams, authentication and project submission flows, a full presentation deck, a working set of feedback agents (Feasibility Analysis, Scope Definition, Tech Stack, Timeline, and Risk & Mitigation), and a one-on-one chatbot. One milestone remaining before the internship concludes.

PROJECTS

PhysioShift — Cross-Hardware Robust Representation Learning for Physiological Signals *(AI/ML/DL/CV, Team Project, In Progress)*

- Self-supervised representation learning framework for physiological time-series signals (ECG, PPG, GSR, skin temperature), built to be robust across sensor hardware domains rather than just across patients or datasets.
- Two applications are being built on top of the learned representations: VitalDrift (arm-worn autonomic health-drift alerting) and Follicular (health-tech classification via pretrained embeddings).

PhaseSAR-Net — Phase-Aware SAR Intelligence for Robust Scene Classification *(AI/ML/DL, Solo Research, In Progress)*

- Built a phase-aware (complex-valued) SAR classification system using paired real/imaginary convolutions (ComplexConv2d, ComplexBatchNorm2d, ModReLU) and benchmarked it against a magnitude-only baseline CNN (~490K vs ~423K parameters) to study AI robustness under sensor-realistic corruptions.
- Designed a physics-inspired phase-consistency regularization loss (KL-divergence between clean and phase-corrupted predictions) combined with cross-entropy, reducing baseline accuracy degradation under speckle, Gaussian, and platform-jitter corruptions.

LakeGuard AI + AquaDrone Sentinel *(Team Project)*

- AI sentinel fusing live IoT sensor swarms with ML to catch water-quality anomalies early in Karnataka's lakes; placed 4th of 147 teams at the KSRSAC State-Level Competition (Govt. of Karnataka).

ACHIEVEMENTS

- **4th Place of 147 teams, KSRSAC State-Level Student Innovation Exhibition** (Govt. of Karnataka) for LakeGuard AI, winning ■15,000
- **250+ problems solved on LeetCode**, maintained through a sustained daily-practice streak across DSA topics
- **300+ day streak on Duolingo**
- **Active hackathon participant** across multiple Bengaluru-area college hackathons

LANGUAGES & INTERESTS

- **Languages:** English, Kannada, Hindi, Japanese
- **Interests:** AI-driven side projects, hackathons, travel, comics